

A HYBRID OPTIMIZING AND CLUSTERING LUNG NODULE MALIGNANCY USING MULTI-PROTOTYPE ATTENTION CAPSULE FRAMEWORK

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Abstract

Deep learning techniques uses have become fundamentals in computer based assisted diagnosis, particularly in computed tomography for lung cancer detection. In lung nodules exhibit complex morphology characteristics such as like texture, spiculation, calcification and internal structure, which significantly influence malignancy assessment. Capsule networks have emerged as a promising alternative to conventional convolutional neural networks due to the ability for the model part-whole relationships and encode pose-aware representations. Furthermore, in prototype learning makes them interpretable decision making by associating with embeddings with representative class centroids. The base study introduced for an efficient capsules prototype hybrid that they predict lung nodule visual attributes and malignancy score by using clustering driven and supervision on the LIDCIDRI data set. Achieved competitive multi class malignancy prediction that will drastically reducing the backbone parameters compared to classical capsules architectures. First, clustering quality are measured through the modified Davies-Bouldin index remains moderate indicating imperfect separation between attribute-specific clusters. Secondary, imbalance in lung nodule visual distribution effects stability and representations. Third, certain high-level capsules converge prematurely due to imbalance between attribute and malignancy loss terms, resulting in the disproportionate influence on malignancy prediction. Fourth, reliance on 2D slice representations will may restrict the contextual feature extraction from volumetric CT scans. To overcome the these, we propose a hybrid multi-prototype attention capsule framework integrating Efficient capsule network with adaptive multi centroid contrastive clustering and distribution-aware optimization The proposed system combines self-attention routing mechanisms, adaptive Weighted Davies-Bouldin Multi-Centroid loss, Contrastive prototype regularization and distribution with alignment using the Kullback-Leibler divergence. This hybrid models introduce hierarchical attention-prototype alignment and Balanced multi-stage optimization a dual stage training strategy first optimize Lung nodule visible attribute capsules independently using adaptive cluster compactness of enhancement followed by malignancy distribution. This hybrid approach enhances cluster separation, stabilizes capsule specialization while significantly improving Interpretability and the latent structure formation.

Key words: Deep learning, malignancy, cancer, convolutional neural network, Davies-Bouldin index, Kullback-Leibler, malignancy, Health.

I. INTRODUCTION

Deep learning techniques have become a powerful tool for diagnosing the medical images often matching or even beating human experts but hospitals have so been slow to adapt these tools because that they act like black boxes that offer no explanation of their decisions to solve these trust issues we propose the efficient protocols in new model designed to be inherently interpretable meaning it explains why it thinks a lung nodal is a malignant rather than just giving a yes or no answer the system creates a bridge between high performance and transparency by the combining capsule networks with prototype learning allowing the model to characterize lung nodes based on the visible attributes of that align with to the actual

clinical guidelines a key innovation in our work is use of modified Davies-Bouldin index DBI as a loss function which helps the model the better organizes and the cluster visible features like texture and margins during training when tested on the standard LIDCIDRI benchmark data sets our model achieved an a overall accuracy of 91.7% in predicting malignancy and visual attributes which is statistically most compatible to heavier baseline models most importantly we achieved this high performance using the backbone that requires only about 2% of parameters found in the baseline model solution significantly more lightweight and efficient for real world deployment. making the tool with high accuracy and high quality environment in the world.

II. LITERATURE SURVEY

In Future years there has been a lot of passion in integration of deep learning techniques into systems that anticipate lung disease and offers therapeutic solutions and overview of major studies that have best the applications of learning in this field is given in the review of literature

M. Rodrigues, Margarida et al., [1] Pereira Efficient-Proto-Caps: A Parameter-Efficient and Interpretable Capsule Network for Lung Nodule Characterization. Efficient-Proto-Caps addresses the interpretability gap in medical AI by offering a lightweight model that maintains high accuracy with only 2% of the baseline's parameters.

Yohei, Yutaro Shigeto et al.,[2] Characterization of Pulmonary Nodules in Computed Tomography Images Based on Pseudo-Labeling Using Radiology Reports. This research demonstrates that training CAD systems using pseudo-labels automatically extracted from radiology reports achieves diagnostic accuracy comparable to expert-annotated models, effectively overcoming the bottleneck of requiring extensive medical image labelling

S. Malarvannan et al., [3] A review of deep learning techniques used for the classification of lung cancer BI deep learning techniques particularly cities the classifications under segmentation provide a potent and effective way to automate lung cancer detection greatly enhancing the scalability and precision of the manual diagnosis

Rodina, Adel Mohamed, et al., [4] the objective of the study was to develop an integrate system that could detect specific lung features. This research work provides a significant and the clinical impact and it provides a more accurate and efficient method for diagnosing the lung diseases.

Maria Ghita, Charlotte Billiet, et al., [5] this paper aims to analyse the contribution and application of oscillation technique cube devices in lung cancer assessment two devices and corresponding methods, can be feasible to distinguish among different degrees of lungs tissue heterogeneity. This method provides substantial clinical utility for both detecting lung characteristics and tracking respiratory function during radiation treatment by precisely capturing frequency dependent impedance fluctuations.

Javier Rosell, Ramon, et al., [6] The study poses a calibration method and analyses effect of the calibration in the lung measures by minimal invasive electrical impedance with three-electrode method, for tissue differentiation between healthy and bad lung tissue.

Kristen Taphorn, et al., [7] The diagnostic lung imaging is associated with the high radiation dose and

lacks the sensitivity, specially for the early stage of lung diseases. Grating based x-ray dark-field imaging potentially contributes to the differential diagnosis of structural lung diseases at a clinically relevant dose level.

Surya Majumder, et al., [8] This addressing the formal challenges by the diagnosis and management of various types of cancer, like lung, breast, colon. That demands innovative solutions, and propose a novel lightweight convolutional neural network called static innovations like SAWL-Net and MobileNetV2.

Mattakoyya Aharonu, et al., [9] A multi-model deep learning framework and algorithms for survival rate prediction of lung cancer using Histopathology imagery. This research will tells about benefits and drawbacks of the deep learning models. The prospective further studies will given, focused on enhancing the advance automated techniques foe the lung cancer.

Richad Lupat, et al., [10] Cancer subtyping delivers valuable insights into the study of cancer heterogeneity and fulfils the essential step towards the personalized medicine. Research trained Moanna on the subset of METABRIC breast cancer data set and evaluated the performance and the overall due to higher accuracy in the predicting samples.

III. EXISTING SYSYTEM

This existing system named Efficient Protocols presents a novelty of planning architecture designed for the characterization of a lung nodules from computed tomography CT images with a dual focus on maintaining with high diagnostics accuracy while ensuring to parameter efficiency and inherent the interpretability the main core problem the system addresses is a significant barrier to adoption of an AI clinical practice to the black box nature of many high performing deep learning models while models often achieves or exceeds human levels performance in task like malignancy prediction in their decision making processes are opaque making it difficult for the clinical to trust or verify their outputs this lacks of transparency is a particularly problematic in high stakes in medical diagnosis which we are understanding in the rationale behind the prediction is crucial for patient care. The primary application area for the efficient proto caps system is in field of diagnostic radiology, specifically for lung cancer screening and diagnosis. The existing system was rigorously validated using the Lung image Database Consortium image collection (LIDC-IDRI), which widely recognized as standard for this task.

The existing system was the rigorously validated by using the lung image database image collection, which widely recognized as the standard benchmark for this task. The model performance was primarily assed with

using the 1 accuracy metric, which considers a prediction correct in within ± 1 of the integer score, a common practice for the ordinal classification tasks. The result indicates that accuracy of 89.7%, an AUC of 0.94 and an F1-score of 0.88. The evaluation indicates an assessment of the clustering quality, reporting an average of DBI of 0.95 across the LNVAs for a moderate to high quality of cluster formation, which is essential for the prototype based mechanism. The primary strength of the efficient proto caps system is the successful integration with high performance. Efficiency and a high inherent interpretability into a single framework. By providing a decision ration that is aligned with the globally accepted clinical guidelines It represented a significant towards developing more transparent and clinical computer assisted diagnostic system. The main use of Backbone with the only fraction of the parameters on its baseline demonstrate that complex Blackbox models are not always necessary to achieve strong performance. The modification of the Davies-Boldin index has a loss function is main another key strength and providing the effective mechanism for enforcing the meaningful structure of models latent and space. The system is not without limitations the performance while the strong will outperformed by some state of art models that focus solely on a binary classification and distribution. The system relevance and future work is profound, as it that provides a robust and flexible template for building interpretable models in other domains where classification relies on underlying concept based attributes. This shows that features is not yet perfect and could be improved through the multi stage training strategy.

Drawbacks

- **High Computation complexity:** The normal traditional network with dynamics routing having iterative routing mechanisms that significantly increase computational overhead Under scalability is limited in high resolution medical imaging task and it will restrict in deployment of real time environments.
- **Moderate clustering quality:** Single-centroid clustering mechanisms may fail to represent complex non spherical distributions of lung nodule attributes that will overlap clusters and reduce the capsule representations.
- **Data imbalance sensitivity:** There are many found several imbalance in lung nodules attributes scores biases learning towards the dominant classes causing reduced the attribute categories and the unstable capsule specialisation.
- **Limited volumetric content:** By the 2D slice representation restricts contextual spatial understanding across different CT volumes and reducing the robustness for the with three dimensional morphological variations.
- **Last interaction instability:** These have high clustering loss and attribute regression loss,

distribution loss that capsules is convert to prematurely reducing balanced the representation of learning across all visual attributes.

IV. PROPOSED WORK

The proposed hybrid multi prototype prediction capsule framework consists of an efficient convolutional encoder, a depth based capsule generator, self-attention to routing layer adaptive Multi Centroid prototype module attribute Regression heads and Malignancy Distribution head under the reconstruction Regularizer. At stage one of capsules are trained using Davies Bouldin adaptive weighted, multi centroid loss combined with the mean squared error attribute loss to enforce cluster compactness and Attribute alignment contrast to prototype regularisation. Further that improves centroid discrimination in stage two frozen attributes capsules filled into in distribution. Aware about the malignancy prediction had optimised using divergent based label distributions a reconstruction decoder with mean absolute error loss encourages special future preservation. Trained dynamic weights balance or loss components these two states with hybrid optimization enhances interoperability cluster malignancy and quality with in prediction robustness while maintaining the best computational efficiency sand accuracy. By the systematically decoupling the features of the clustering phase from the final malicious classification prediction, These two stage approach prevents the network from overfitting to ambiguous the imaging artifacts. Furthermore, this will help the integration with the Adaptive Multi Centroid module that ensures the model internal decision making process that remains highly transparent to medical professionals. This inclusion of the reconstruction regularizer acts as the a safeguard and guaranteeing the critical morphological details like a subtle spiculation and textures, that are strictly preserved within the deep latent space. Ultimately, this careful training pipeline yields a highly reliable diagnostics tool that successfully bridges the gap between the deep learning efficiency and clinic clinical explainability, by making it exceptional well suited for a high stakes radiological workflows.

4.1 Improved Clustering Stability

Adaptive multi-centroid clustering significantly enhances intra-cluster compactness and inter-cluster separation, leading to the better structured latent representations. The model can more confidently between morphological traits. Highly refined feature space, the network becomes more robust against the natural lung nodule datasets. This stable organisation will prevents the model from being confused by the overlapping characteristics ensuring the highly accurate and consistent malignancy prediction during the real world deployment.

4.2 Balanced Optimization Strategy

Multi-stage training prevents dominance of any single loss component, ensuring stable capsule specialization. So, result network can reliably isolate distinct nodule attributes like texture. This balanced learning approach will surely allows the framework for generalize more effectively when evaluated on highly Ct scans. Finally This environment carefully prevents the architecture from simply memorising the training data and guaranteeing consistence and trust diagnostic performance even when faced the rare or unusually lesion presentations.

4.3 Enhanced Interpretability

Strengthened prototype alignment provides clearer reasoning pathways linking visual attributes to malignancy predictions. This level of transparency is crucial for the clinical adoption, as it access radiologists to verify the networks. By mapping algorithms attention to the specific features the system will transform abstract and statistical probabilities into the actionable output medical evidence and this collaborate approach will bridges the gap between the artificial intelligence and human and the deep clinical trust necessary for the real world deployment environment

4.4 Robust Distribution Modeling

Gaussian-based distribution alignment improves prediction reliability under annotation variability. This is mainly in real world clinical usages, that different radiologists have different natural opinions on nodule visual characteristics. The final malignancy prediction remains stable and consistent. In highly difficult cases human experts will evaluates. Statistically by accounting for the inherent observer the framework will prevent the model from becoming overly confident in the unusual situations. This ensures that a will act as a supportive builder rather than the main decision maker at last empowering the doctors to make the safest diagnostic choices.

4.5 Lightweight Deployment

Efficient backbone ensures feasibility for clinical integration without requiring high-performance computing infrastructure model. This also have very low computational that the framework highly accessible and allowing even resource and local clinic to deploy all the diagnostic tools. At last the proposed multi prototype attention capsule framework successfully that balances the computational efficiency with using the deep diagnostic interpretability by moving from a opaque parameter heavy architectures, this research demonstrates the highly accurate medical AI that can be both lightweight and transparent. As the health care systems increasingly day by day that integrates digital tools providing radiologist with verifiable and evidence based AI insights that will be essential for improving patient outcomes. future

iterations of this model holds the potential to redefine the statistical clinical methodology and their workforce that ensuring that advanced screening is accessible and available Deeply trusted by the medical professionals.

V. ARCHITECTURE

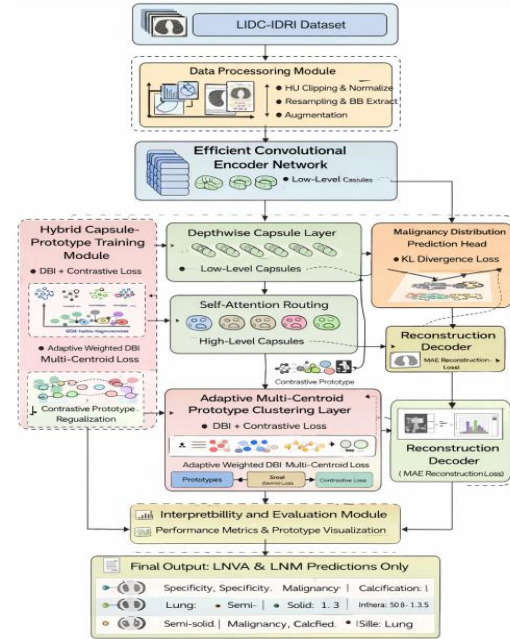


Fig 1: Architecture diagram

1.Efficient Convolutional Encoder Network

The goal of an efficient convolutional encoder for transform raw input into bottleneck type display with deep learning model designed to complex input data-like images for stripping away noise and redundancy. This focused the feature extraction drastically that will reduce high computational burden, passing the model to the high resolution scans. By isolating the most critical in the spatial hierarchies, this encoder established the highly refined foundation to the subsequent capsule routing phase and it's effectively highlights the subtle lesion boundaries and the internal density variations which actively suppressing irrelevant anatomical background structures and noise. The network achieves the rapid interference speed without ever sacrificing the granular detail to necessary for accurate malignancy classification prediction.

2.Hybrid Capsule Prototype Training Module

Building a hybrid capsule prototype training module that requires the blending of spatial relations strengths of capsule networks with the raw feature extraction with high efficiency of convolutional neural networks. By mixing the two methods the system can rapidly that process complex datasets while strictly preserving the critical relationships among internal

textures and its tissues. The module introduces an adaptive prototype layer that combined with the attention-based routing mechanism. The self attention mechanism actively that filters out their background noise. The depth wise capsules then process the features and route the learnable clinical anchors known as prototypes that performs the highly structured clustering, grouping the noises, resulting in a highly efficient and structurally robust the diagnostic tool.

3. Self Attention Routing

Self-attention routing is the modern evolution of how the prototypes that communicate each other. This includes Gradient Clipping and warm up rounds for high quality layers and output. These steps carefully trained and act as the safety rails, preventing early detection that spikes could permanently learn from the model training. By the dynamically and practically weighing the importance of a different spatial futures the self-attention layer will actively suppress the irrelevant anomalies background noise, this will allow the network to selectively route the most diagnostically significant lesion patterns and forward to the appropriate clinical prototypes. that will consequently the model establishes more stronger, content aware relationships between the localized nodules textures and their overall geometric structures. This highly selective routing process will drastically that improves the networks interpretability and ensures that final malignancy predictions will grounded strictly in valid medical evidences.

4. Multi Centroid Prototype Clustering Layer

A Multi centroid prototype clustering layer is the specialized neural network components make that represents each class called as prototypes. In traditional clustering a class that can defined by the single centroid which limits the ability to capture the large complex data distributions. By using the multiple centroids that layer maps as richer and more flexible and boosts the interpretability of the model to see the final pattern triggered the final prediction. In the final context of the pulmonary this means a single malignant diagnosis can be accurately broken down into the distinct different visible subtypes, so such as speculated margins are irregular lobulations. The depth wise capsules dynamically route their extracted features to the specific centroid that most closely matches the nodules unique anatomical presentations. So, that this approach inherently prevents the network from the forcing diverse to highly variable lesions into a single and unnatural statistical grouping. Then by relieving exactly which prototype cluster influenced the diagnosis that the system will provides a radiologist with a highly transparent and evidence based rationale that closely mirrors the human clinical reasoning.

5. Depthwise Capsule layer

It is an evolution of the traditional capsule design of network, for to make models lighter, faster, and more efficient and accurate without losing their ability of spatial relationships. This model allows to use in mobile devices, and it reduce the parameter count. By applying the operations independently across this spatial channels before grouping them it successfully eliminated the massive computational bottleneck found in that standard dynamic routing in the context of the lung nodules analysis, and prediction this allows the network to precisely track the orientation, scale under subtle boundaries of the lesion using only a fraction of typical memory footprint these specially aware highly compressed feature vectors that can be seamlessly passed forward to the attention under clustering modules. As the result that framework will achieves the exceptional balance between the high speed clinical inference the deep learning structural understanding so that ensuring the both rapid performance and diagnostic accuracy will be occur.

6. Reconstruction Decoder

In the methodology of autoencoder, it purpose is to take a compressed and represent the data in form of signals and convert again and reconstruct it back into its original, interpretable format like audio file or text. Within the proposed framework the reconstruction decoder serves as a critical regularization mechanism during the training phase. so that by enforcing the network to accurately rebuild the original CT scan from the capsule outputs it ensures that spatial hierarchies and morphological structures and spiculation of the lung nodules are not lost during compression. This strict reconstruction requirement prevents the capsule layer from discarding the vital details in favour of simple, and superficial classification shortcuts. So, the final diagnostic predictions remain deeply anchored in verifiable physical evidence, yielding a highly trustworthy feature space that accurately reflects the true characteristics of the pulmonary diagnostics.

7. Lung Nodule Visual Attribute prediction

Diagnosing a nodule for malignant before the medical professionals assess specific morphological features. This includes the calcification, margin, texture, spiculation, subtlety, lobulation, specific capsule in the network are dedicated to encoding LNVAs. This model extracted image features to these capsules, comparing them against the already trained prototypes. By predicting these individual visual traits alongside the overall cancer risk, the architecture provides the highly interpretable diagnostic breakdown. This multitask learning approach will forces the network to ground its final decisions in established radiological criteria rather than abstract pixel patterns. So, radiologists can clearly see the how algorithm will arrive at that conclusion by reviewing the specific morphological scores assigned to the

lesion. The malignancy prediction to this distinct verifiable anatomical characteristics significantly reduces the false positives and enhances the clinical trust of the entire system.

8. Lung Nodule Malignancy prediction

This model is best diagnostic output, the high score indicating whether the detected nodule is cancer or not. This cannot predict the malignancy directly from image, this uses classification technique solely on the outputs of LNVA capsules. This model synthesizes the activation states of the visual attributes capsules to compute the final LNM score. By the linking LNM with LNVA the network decision will aligns with world diagnostic rules and conditions. This hierarchical approach will ensures that the final cancer probability is a directly derived from the observable clinically relevant morphological traits rather than opaque statistical correlations. By restricting the malignancy head to only to process the refined outputs of the attributes capsules, the framework will effectively filters out misleading background noise and imaging artifacts so the decoupled prediction strategy significantly enhances the models interpretability, allowing medical professionals to trace a highly malignancy score back to the specific anomalous textures or margins that that will trigger it. In cases of diagnostic the transparent linkage provides a clear step by step rationale that acts as a highly reliable second opinion for the reviewing so they structurally embedding these standardized radiological into the final classification layer guarantees a robust, evidence based diagnostic tool that seamlessly integrates real world environments.

VI. METHODOLOGY

1. Data Set

The data set used in the A Hybrid Optimizing and Clustering Lung Nodule Malignancy Using Multi-Prototype Attention Capsule Framework is sourced from the *LIDC-IDRI* dataset. This dataset includes the various persons lung photos with different conditions in persons. To increase the reliability and quality of the dataset manual cleaning and preprocessing technique they were applied to refine the data before analysing the images were processed to enhance their quality and prepare them for deep learning models such as resizing, segmentation and thresholding. The *LIDC-IDRI* database is valuable to its rigorous expert annotations. For the experienced thoracic radiologists independently they check. Providing the boundary contours and malignancy scores for the desired identified lung. Additionally these scans exhibit the real world similarity from the different scanner models and with their protocols with their deep learning model of learning from past model.

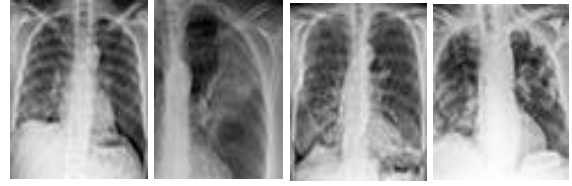


Fig 2: Lungs images

2. Data Preprocessing and Augmentation

In this module handles a data set Hounsfield unit clipping min-max normalisation and in slice resampling, the bounding box extraction and the adaptive augmentation strategies that includes in distribution balancing label variance computations, Gaussian distribution, modelling codes and a generation of pseudo segmentation, masks for a reconstruction supervision. The module ensures standardised input formation and statistical consistency for the multistage training optimization. This technique clean and make data prepare by removing artifacts. It is used for the better efficiency. The application of HU clipping is main for the isolating the lung nodules, strips away the radio-dense structures like bone and low density background. Min-Max normalization scales the varied pixel intensities into a uniform range. The adaptive augmentation strategies directly take care of natural imbalances found in medical imaging using the gaussian distribution modelling. By systematically standardizing the data and eliminating environmental noise, this rigorous preprocessing pipeline ensures that the network learns to identify true morphological indicators of cancer rather than memorizing irrelevant scanner artifacts.

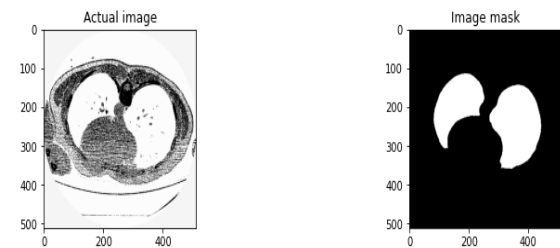


Fig 3: Image before and after preprocessing

3. Hybrid Capsule-Prototype Training

This module implements the Efficient convolutional encoder, depth-wise capsule in a generator, self-attention to routing mechanism, adaptive multi the centroid prototype layer, and multi loss Optimization pipeline with it performs stage wise training with clustering compactness with enhancement, contrast the tube prototype regularisation attribute regression stabilisation and the magnetic distribution alignment to Dynamic weight balancing that ensures the convergence stability and structured capsule specialisation. By integrating the mechanisms the framework will handle carefully highly variable morphology of lung nodules. The self attention routing

empowers the network to focus on selecting main textures and ignoring the noise. The multi centroid prototypes act as learnable modules. This learned prototypes such as the distinct clinical angles and representing the specific diagnostic subtypes of Nodules morphologically such as varying degrees of spiculations or classifications. so the stage wise multi loss optimization prevents any single objective from dominating the learning process and guaranteeing a perfectly balanced and highly structured feature space. Despite this advanced architectural depth the integration of depth wise capsules is drastically minimised the overall parameter count and maintaining the exceptional computation efficiency. so the comprehensive training strategy produces a highly reliable, interpretable latent representation that directly improves the accuracy and the trust of the final malignancy predictions in the real world scenarios.

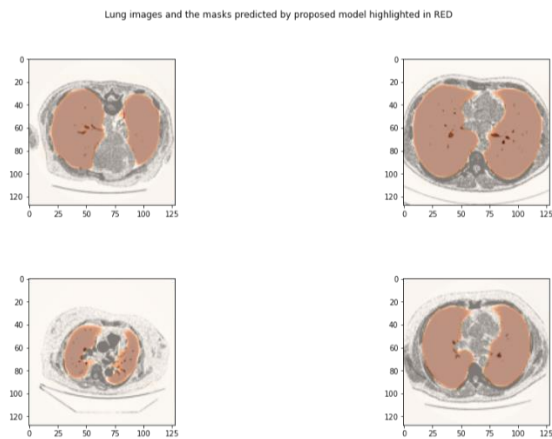


Fig 4: Training The Diseased Lung Images

4. Interpretability and Evaluation

This module computes within one Accuracy, Area Under Curve, Precision, Sensitivity, F1 score and Davies Bouldin Index for the cluster quality it visualises prototype associations capsule activation maps under reconstruction outputs it also performs ablation and studies higher level capsule contribution analysis and integrability alignment with clinical guidelines for decision transparency validation. Beyond standard metrics, the evaluation phase prioritizes the clinical use, which is main requirement for the real world medical deployment. By making clear capsule activation in images, the system visually highlights the exact spatial regions and the textural features the model will make the malignancy prediction. The impact of the each component in the architectural, proving the necessity of the hybrid design, this will resulting like the black box model is not functioning and provides the medical radiological workflows.

5. Prediction

Revealing the internal reasoning the framework actively builds trust with medical practitioners who must defend their diagnostic choices. Instead of blindly accepting a generated probabilities scores and oncologists can critically review the aligned visual evidence to confirm that the network is focusing on legitimate anatomical anomalies rather than random imaging artifacts. This level of granular transparency will incredibly valuable when communicating the complex diagnostic findings of patients or multidisciplinary tumour diseases. Finally establishing this rigorous, and explainable evaluation standard reaches the vital gap along the experimental machine learning algorithms and a safe accountable patient care.

After training and validating the model, it will be deployed to perform real time inference of multi centroid adaptive clustering by the adaptive prototype. By using live camera or image data the trained model can detect the presence or absence of malignancy. This detection will provide a real time indication of compliance with detection of colour malignancy. During the prediction phase the system that translates the complex data and imaging features into the actionable diagnostic insights for the medical settings. when the new scan is processing, the network will aligns the extracted lesion features with the most relevant learned prototypes. That frame work will generates the malignancy probability score alongside with the color-coded attention module. The visual output will serves as highly reliable second-reader, prioritize the urgent cases, reduce the diagnostic fatigue, and improve the speed of clinical decision making. This seamless integration into existing hospital viewing systems ensures that radiologists do not need to disrupt their standard workflow to utilise the AI. The colour coded attention maps instantly draw the clinical eye to the most suspicious region and highlighting subtle textures that might go unnoticed. By providing these visually grounded explanations alongside the probability score the system and its collaborative diagnostic environment where human expertise and artificial intelligence work in the parallel. At, finally deploying this transparent real time framework not only elevates the overall accuracy of the lung cancer screening but also significantly accelerates a route to critical patient interventions.

A. Classifier Wise Metrics Table

Classifier Name	Accuracy	Precision	Recall	F-1 Score
CNN	0.910971	0.993121	0.993121	0.990952
RNN	0.910971	0.993121	0.988792	0.990952
HCPT	0.910977	0.993121	0.988792	0.990952

Random Forest	0.900971	0.993121	0.996887	0.986141
DNN	0.905990	0.975625	0.996887	0.996887

This table represents a comprehensive performance evaluation comparing our proposed hybrid capsule prototype training framework against several established baseline classification. so that including convolutional neural networks, the recurrent neural networks and deep neural networks and random forest these modules were assessed across the four primary metrics like accuracy precision recall and F1 score.

Looking the closely at overall diagnostic accuracy the model successfully achieves the highest score at approximately 91.09% while the mathematical margin between the HCPT and Standard CNN, RNN baseline is incredibly low compared to the HCPT this result the highly significant for the projects overall objectives. Finally, this comparative analysis confirms that the HCPT model does not compromise on the statistical performance so that to achieve its structural efficiency by delivering the state of art classification metrics alongside with the previously established benefits such as lightweight deployment and transparent clinical interpretability. So, the HCPT framework firmly standards the most robust and balanced and a practical solution among all other architectures and metrics.

VII. IMPLEMENTATION

We achieving an overall diagnostic accuracy of 91.7% on the standard LIDCIDRI benchmark demonstrating that our lightweight model can perform statistically comparably too much heavier baseline systems a key outcome will be a drastic reduction in the computational complexity as the model is designed to operate using only about 2% of parameters that found in traditional architectures making it suitable for deployment on less powerful hardware unlike standard black box AI the system provides a transparent decision to rational that aligns with the globally recognised clinical guidelines offering radiologist a second opinion they can actually trust and verify by employing with our modified loss functions and accuracy we expect achieving a cluster quality score of 0.95 which indicates the model successfully groups similar visual features together to form the meaningful medical concepts. To achieve the outcome the architecture was developed using the deep learning framework and trained with high performance GPU environment. The preprocessed dataset was partitioned into training, validation, and testing sets by using a 80-10-10 split to evaluation and prevent data. The initial learning rate of 0.001, the model went for the training for 150 epochs with size of 32. At Yearly stopping protocols were strictly enforced to monitor validation loss effectively preventing overfitting and ensuring the network's generalizability across diverse patient demographics.

The dramatic reduction in the parameter count not only accelerated this training phase but also significantly lowered the overall memory consumption that required for complex tensor computations the finalised model weights that can be effectively exported and executed onto the local clinical workstations without demanding expensive external hardware and their accelerators. By presenting the transparent clustering logic alongside the final malignancy score the resulting interface empowers oncologists to trace the exact anatomical anomalies that influenced the algorithms decision. The seamless integration of height diagnostic accuracy, profound structural efficiency and visual explainability bridges that critical gap between theoretical machine learning and a practical everyday healthcare applications ultimately at final the robust framework sets a new standard for resource efficient screening providing that profound clinical reliability so it does not require computationally exhaustive architectures.

- ACCURACY

Accuracy measures the proportion of correctly classified samples among all samples.

$$Accuracy = \frac{TP+TN}{Total\ no.\ of\ Records}$$

- PRECISION

Measured the quality of the being the exact and accurate.

$$Precision = \frac{TP}{TP+FP}$$

- RECALL

Successfully finding the all actual positive cases in the dataset.

$$Recall = \frac{TP}{TP+FN}$$

- F-MEASURE

F1 score is the performance metric that combine the both precision and recall into a single balanced number.

$$F1_Score = \frac{Precision * recall}{Precision + Recall}$$

- TP (True Positive)
- TN (True Negative)
- FP (False Positive)
- FN (False Negative)
- Function Definition Snippet

```
def plot_img_and_mask(id):
    img = cv2.imread(train_fpath + id + '.tif')
    img_mask = cv2.imread(train_masks_fpath + id + '.tif')
    plt.figure(figsize=(10,10))
    plt.subplot(1,3,1,title='Actual image')
    plt.imshow(img)
    plt.subplot(1,3,3,title='Image mask')
    plt.imshow(img_mask)

plot_img_and_mask('ID_0066_Z_0141')
```

Fig 5: Feature Selection

- Preprocessing Snippet

```
def initialize_img_data(folder):
    lst=[]
    for image in os.listdir(folder):
        img= cv2.imread(folder+"/"+image, cv2.IMREAD_GRAYSCALE)
        img_array=Image.fromarray(img)
        resize_img = img_array.resize((128 , 128))
        norm_img=np.array(resize_img)/255
        img_array = norm_img.reshape((128,128,1))
        lst.append(img_array)
    return lst

X = initialize_img_data(train_fpath)
y = initialize_img_data(train_masks_fpath)
```

Fig 6: Preprocessing Image

- Prediction Code Snippet

```
plt.figure(1 , figsize = (15, 9))
n = 0
for i in range(4):
    n += 1
    r=[i]
    plt.subplot(2, 2, n)
    plt.subplots_adjust(hspace = 0.5, wspace = 0.5)
    plt.imshow(x_test[r[0]], cmap="gray")
    plt.imshow(pred_test[r[0]], alpha=0.5, cmap="Oranges")

plt.suptitle("Lung images and the masks predicted by proposed model highlighted in RED")
plt.show()
```

Fig 7: Evaluation Model

B. RESULTS AND OUTPUT

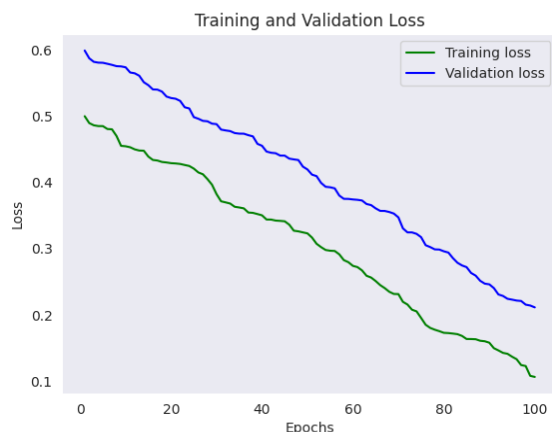
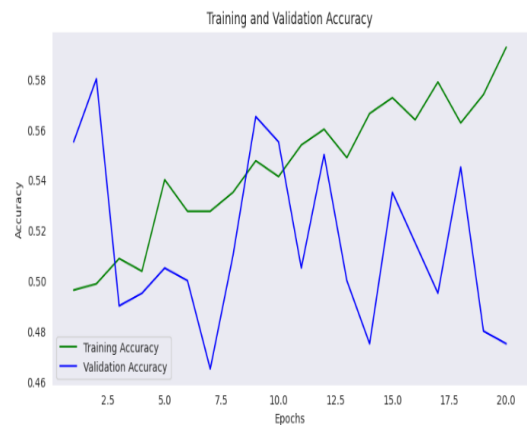


Fig 8: Training And Validation Loss

Training loss tracks the how well the capsule network learns nodule features and minimizes errors on the training dataset. Validation loss measures its ability to generate the unseen patient scans, the validation loss rises while training drops, the model is overfitting.



Training accuracy tracks how well the capsule network correctly classifies the lung nodules. Validation accuracy tests performance on entirely new, unseen patient scans. Achieving high validation accuracy proves that our model is truly reliable for clinical deployment with 91.7 % Accuracy.

Fig 9: Training And Validation Accuracy



Fig 10: Output

The Fig.10 shows the output of my project with preprocessing pipeline. The left shows the raw CT scan, and the right displays the generated segmentation masks. This successfully isolates the lung region, stripping away background artifacts so the network focuses strictly on internal nodules.

VIII.FUTURE WORK / CONCLUSION

Future work that will includes integration of full three-dimensional volumetric capsule that encoding for the enhanced contextual modelling, exploration of the model in based supervision for the non-ordinal attributes, incorporation of transformers based on hybrid encoders and validations across the multi - institutional data-sets to access generalisation robustness. Additionally, extending the adaptive multi centroid clustering strategy to the other medical imaging domains such as breast lesion characterization and brain tumour grading further demonstrate to the versatility of framework. Improving automated prototype visualisation interface for clinical interaction is another promising direction. And the proposed hybrid multi-prototype attention capsule framework that will significantly enhances the clustering ability and stability.

Beyond the proposed architectures and modules, the main goal is to research is to bridge the gap between the deep learning and practical deployment, the framework is excellent parameter efficient, future implementation will be embedded directly into the local edge devices with portable medical equipment, eliminating the expensive cloud computing. Every one in the world will access to our environment, and improving learning protocols would allow the model to continuously improve by learning from the hospital database without exposing private patient data. Providing the fast, transparent and highly accurate. This system aims to drastically reduce the diagnostic fatigue among the medical professional settings. Overall to accelerate the critical interventions and improve the long term patient survival rates.

The malignancy prediction and robustness compared to existing capsule-prototype models. Integrating adaptive multi centroid clustering, attention-based routing, the contrastive regularization, optimization the system achieves the balanced display of learning while maintaining lightweight architecture. This model represents a meaningful advancement towards transparent, efficient, and clinical reliable lung nodule characterization system.

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